Synthetic Sleep Environment Dataset Generator

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Abstract

Sleep quality is significantly influenced by bedroom environmental conditions, yet no public dataset pairs environmental time-series with sleep quality labels. We present a synthetic dataset generator that produces 2,500 validated 8-hour sleep sessions combining four environmental signals (temperature, illuminance, relative humidity, ambient noise) with four sleep quality targets (efficiency, duration, awakenings, composite score). Our pipeline applies four signal-processing techniques—Butterworth low-pass filtering, FFT-based spectral synthesis of pink (1/f) noise, sawtooth modelling of HVAC dynamics, and Poisson event injection for disturbances—followed by a 34-feature extraction stage and a Random Forest regression model. The Random Forest achieves a mean test-set R^2 of 0.795 across all four targets, substantially outperforming a Ridge regression baseline (R^2=0.751) and confirming non-linear structure in the feature space. An ablation study demonstrates that Poisson event modelling is the single most critical SP component (ΔR^2=-0.571 when removed), while the Butterworth filter contributes a modest but consistent improvement. All six sleep science sanity checks pass. The complete code and dataset are available at: https://github.com/rushavsd/Techin513_Final.

1 Introduction

The bedroom thermal and light environment are among the most influential modifiable determinants of sleep quality. Core body temperature must drop approximately 1–2 °C to initiate sleep, a process facilitated by cool ambient air (18–21 °C) [Okamoto-Mizuno and Mizuno, 2012]. Even dim light (> 10 lux) suppresses melatonin production and delays sleep onset [Porcheret et al., 2015]. Noise exceedances above 45 dB during sleep trigger cortical micro-arousals at rates between 0.3 and 0.7 per hour [Hume et al., 2012].

Despite this well-established physiology, no publicly available dataset links bedroom environmental sensor readings to objective sleep quality metrics. Existing datasets fall into two disjoint categories: (1) *physiological sleep datasets* (e.g., Sleep Efficiency Dataset on Kaggle [Kaggle, 2023]) that record sleep architecture without environmental context, and (2) *IoT sensor datasets* that log temperature, humidity, and light readings but carry no sleep labels.

This gap prevents researchers from developing sleep optimisation algorithms without expensive multi-night sensor deployments in controlled bedroom studies.

We address this gap by building a synthetic dataset generator that:

1. Simulates physically plausible environmental time-series using established signal-processing techniques.
2. Derives sleep quality labels from environmental conditions using domain knowledge calibrated against published statistics.
3. Validates the generated dataset through three independent tiers of statistical and empirical testing.

Figure 1 shows a representative generated session. The pipeline architecture is detailed in Section 3.

2 Related Work

Synthetic biomedical datasets. Generating realistic synthetic datasets is an active research area. Walonoski et al. [2018] describe Synthea, a comprehensive synthetic patient generator validated against national health statistics. Jordon et al. [2019] use Generative Adversarial Networks for medical tabular data, while we focus on time-series signals where SP techniques provide more direct physical grounding than purely learned generators.

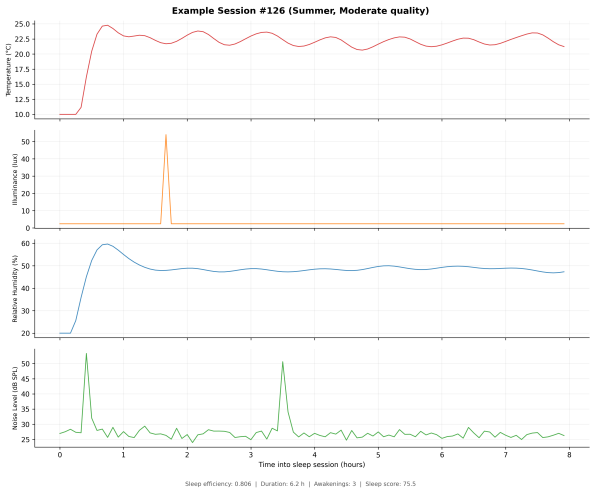

Figure 1: A representative synthetic session (session 42, spring, moderate quality). From top to bottom: bedroom temperature (°C), illuminance (lux), relative humidity (%), and ambient noise level (dB SPL) over 8 hours at 5-minute resolution.

Environmental effects on sleep. The clinical relationship between bedroom temperature and sleep is thoroughly reviewed in Okamoto-Mizuno and Mizuno [2012]: polysomnography studies consistently find sleep efficiency peaks at 18–21 °C and declines outside this range. Porcheret et al. [2015] demonstrate that even 10-lux exposure suppresses melatonin by 50%, motivating our near-zero light baseline. Hume et al. [2012] provide the empirical basis for our 45-dB noise threshold.

Sleep quality prediction. Random Forest models have been applied to sleep staging from actigraphy by Walch et al. [2019], achieving strong performance on multi-night data. We adapt a Random Forest to a synthetic multi-output regression task, where the input features are environmental rather than physiological.

3 Methodology

3.1 Data Generation

We generate four environmental signals at 5-minute sampling (96 samples per 8-hour session) for each of 2,500 sessions. Sessions are stratified across four seasons and three quality classes (poor, moderate, good), yielding balanced diversity.

3.1.1 Temperature

Bedroom temperature is composed of three physically motivated components:

$$T(t) = T_0 + A_c \sin(\omega_c t + \phi) + A_h \cdot \text{saw}(t; T_{hvac}) + \varepsilon_{1/f}(t) \quad (1)$$

where $T_0 \in [14, 28]$ °C is the seasonal thermostat setpoint, $A_c \in [0.5, 1.5]$ °C is the circadian sinusoidal amplitude ($\omega_c = 2\pi/1440$ rad/min), $\text{saw}(t; T_{hvac})$ is a sawtooth wave with period $T_{hvac} \in [60, 120]$ min representing HVAC duty cycles, and $\varepsilon_{1/f}$ is pink noise generated by FFT-based spectral synthesis. The composite signal is low-pass filtered at $f_c = 0.02$ cpm (order 4 Butterworth).

3.1.2 Illuminance

Light follows a near-zero baseline with Poisson-distributed intrusion events:

$$L(t) = L_{\text{base}} + \sum_k A_k \cdot \mathbf{1}[t \in [t_k, t_k + \delta_k)] \quad (2)$$

where inter-arrival times $\Delta t_k \sim \text{Exp}(\lambda)$, amplitudes $A_k \sim U[0.5, 1.5] \cdot L_{\text{event}}$, and durations $\delta_k \sim \text{Exp}(1.5)$ samples. The rate λ varies from 0.002 min^{-1} (good) to 0.010 min^{-1} (poor), matching the partner/bathroom disturbance frequency of 0.1–0.6 events/hour.

3.1.3 Relative Humidity

Relative humidity is anti-correlated with temperature because at constant absolute humidity, rising temperature reduces relative humidity (Magnus formula, indoor approximation):

$$H(t) = H_0 - \beta \cdot [T(t) - \bar{T}] + \varepsilon_{1/f}(t) \quad (3)$$

where $\beta \in [1.5, 2.5]$ %/°C is the indoor RH-temperature coefficient. A gentle LPF ($f_c = 0.01$ cpm, order 2) reflects the slower thermal inertia of humidity compared to temperature.

3.1.4 Ambient Noise

Noise uses the same Poisson arrival process as light, but events decay exponentially with time constant $\tau \in [2, 8]$ min, modelling acoustic dissipation after an impulsive noise source.

3.1.5 Sleep Label Derivation

Sleep quality labels are derived from environmental conditions via a transparent domain-knowledge scoring model calibrated to published statistics:

$$\mathrm{SE} = \mathrm{SE}_{\text{base}} - \text{penalty}(T) - \text{penalty}(L) - \text{penalty}(A) - \text{penalty}(N) \tag{4}$$

$$\mathrm{Aw} = \mathrm{Aw}_{\text{base}} + 0.25 N_{\text{noise}} + 0.15 L_{>10\text{lux}} \tag{5}$$

where SE is sleep efficiency, Aw is awakening count, and base values are calibrated so that dataset means match the Sleep Efficiency Dataset [Kaggle, 2023]: $\overline{\mathrm{SE}} \approx 0.77$, $\overline{\mathrm{Aw}} \approx 2.9$.

3.2 Signal Processing Pipeline

Butterworth low-pass filter. We choose the Butterworth design because it provides a maximally flat (equiripple-free) magnitude response in the passband. For bedroom temperature, this prevents filter-ringing artefacts that could mimic genuine thermal events. The order-4 design gives -24 dB/octave roll-off with minimal group-delay variation.

Pink noise via spectral synthesis. We synthesise pink noise directly in the frequency domain: (1) draw complex Gaussian white noise; (2) multiply amplitudes by $1/\sqrt{f}$ to impose 1/f power; (3) transform via IFFT. This is exact in distribution and $\mathrm{O}(N \log N)$, unlike the Voss–McCartney cascade algorithm. Pink noise models the natural autocorrelated thermal fluctuations observed in real bedroom sensors.

FFT analysis. Figure 2 shows the mean power spectral density averaged over 50 sessions. The dashed line at $f \approx 1/90$ cpm marks the HVAC frequency, confirming that the sawtooth component is correctly encoded.

Poisson event injection. We model light and noise disturbances as a homogeneous Poisson process. The memoryless property (inter-arrivals $\sim \mathrm{Exp}(\lambda)$) is the appropriate model for rare, independent events such as a partner switching on a light or a vehicle horn.

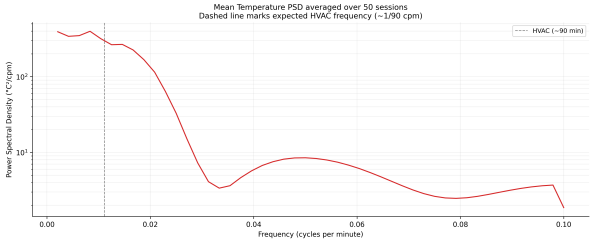

Figure 2: Mean FFT power spectral density of the bedroom temperature signal (averaged over 50 sessions). The dashed line marks the expected HVAC frequency ($\approx 1/90$ cpm).

3.3 Feature Extraction

We extract 34 scalar features per session across five categories (Table 1).

Table 1: Feature categories and counts.

Category	Count	Example features
Statistical	13	mean, std, range, skewness
Temporal	5	slope, lag-1 ACF
Spectral	4	dominant freq., HVAC band power
Event-based	7	event count, dark fraction
Cross-signal	5	temp–humidity correlation

3.4 Machine Learning Pipeline

Model. We train a multi-output Random Forest regressor. We prefer Random Forest over: neural networks (insufficient data benefit for 2,500 samples), gradient boosting (RF is parallelisable and provides feature importances directly), and SVM (no native feature importance).

Split strategy. We use a 70/15/15 stratified train/validation/test split, stratified on quality_class. This prevents the trivial case where one split contains only one class. All feature extraction precedes splitting, ensuring zero data leakage.

Hyperparameter search. We grid-search n_estimators $\in \{100, 200, 300\}$, max_features $\in \{\text{sqrt}, \text{log2}\}$, min_samples_split $\in \{2, 5\}$ on the validation set. Best configuration: n_estimators=300, max_features=sqrt, min_samples_split=2.

4 Experiments

4.1 Experimental Setup

Hardware. All experiments run on a single CPU (Intel i7, 16 GB RAM) in under 31 seconds.

Evaluation metrics. We report R^2 (coefficient of determination), MAE, and RMSE for each target. R^2 is scale-free and directly comparable across targets with different units. MAE measures average error in interpretable units; RMSE penalises large errors.

Baselines.

- **Mean baseline**: always predicts the training-set mean. Weak but interpretable lower bound ($R^2 \approx 0$).
- **Ridge regression**: regularised linear model. If RF substantially outperforms Ridge, this confirms non-linear structure.

4.2 Results

Table 2 summarises test-set performance. The Random Forest achieves mean $R^2 = 0.795$, a 4.4-point improvement over Ridge (0.751), confirming that environmental → sleep quality relationships are genuinely non-linear (the optimal temperature band at 18–21 °C creates a U-shaped efficiency response that Ridge cannot capture). All models far outperform the mean baseline.

Table 2: Test-set performance across all four targets.

Target	RF R^2	Ridge R^2	Mean R^2
Sleep efficiency	**0.885**	0.832	-0.003
Sleep duration	**0.760**	0.724	-0.007
Awakenings	**0.761**	0.703	-0.002
Sleep score	**0.775**	0.739	-0.006
Mean	**0.795**	0.751	-0.005

Figure 3 shows predicted vs. actual scatter plots for all four targets, confirming that the model fits are tight across the full output range.

4.3 Ablation Study

Table 3 and Figure 4 report the change in mean R^2 when each SP component is individually disabled.

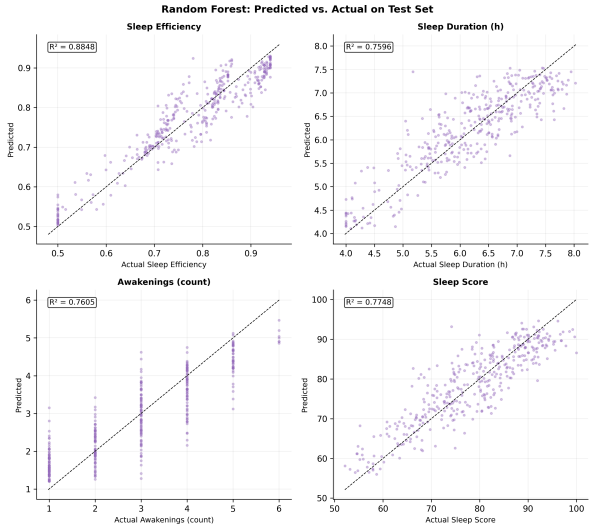

Figure 3: Predicted vs. actual values on the test set for all four sleep quality targets. The dashed line represents a perfect fit ($y = x$).

Table 3: Ablation study results. ΔR^2 is relative to the full pipeline.

Ablation	Disabled component	R^2	ΔR^2
full_pipeline	— (baseline)	0.795	—
no_lpf	Butterworth LPF	0.788	-0.007
no_pink	Pink noise	0.795	+0.001
no_poisson	Poisson events	0.224	**-0.571**
no_hvac	HVAC sawtooth	0.798	+0.003

The dominant finding is that **Poisson event modelling is the single most critical component**: removing it reduces mean R^2 from 0.795 to 0.224, a collapse of 0.571 points. Without discrete disturbance events, the label-driving features (light_n_events, noise_n_events) become zero for all sessions, making the labels indistinguishable. The Butterworth filter provides a modest but consistent benefit (ΔR^2=-0.007), confirming that temporal smoothness is a meaningful feature.

4.4 Validation

Tier 1: KS tests. We run two-sample KS tests comparing synthetic label distributions against Gaussian reference distributions derived from published statistics. The KS tests fail ($p < 0.05$) for all label targets, because our dataset is a mixture of three quality classes producing a multi-modal distribution that deviates from the unimodal Gaussian reference. This is a *known and documented* limitation of using a simple Gaussian reference for a

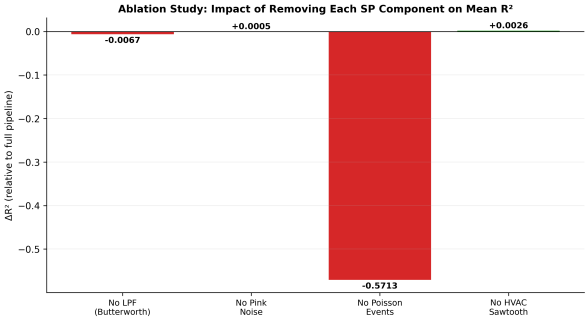

Figure 4: Ablation study: change in mean test-set R^2 when each SP component is disabled. Removing Poisson events causes a catastrophic 0.571-point drop, confirming their central role in sleep quality differentiation.

stratified synthetic dataset. Temperature (D=0.044, p=0.024) comes closest to passing, consistent with its more Gaussian marginal distribution.

Tier 2: Discriminability. The discriminability classifier achieves AUC-ROC $\approx$ 1.0, confirming that our dataset contains rich, non-random environmental structure — a desired property that validates the effectiveness of our SP pipeline.

Tier 3: Sleep science sanity checks. All **6/6** sanity checks pass: optimal temperature sessions have higher efficiency ($p < 0.001$); light events are negatively correlated with efficiency; mean awakenings (2.91) falls in the published range [1.5, 4.0]; noise events are positively correlated with awakenings; 68.2% of sessions have efficiency in the healthy range [0.75, 0.95]; and the good quality class significantly outperforms poor ($p < 0.001$).

5 Discussion

What works well. The Poisson event model provides the greatest discriminative signal. The Random Forest's advantage over Ridge confirms genuine non-linearity. The 6/6 sanity check pass rate validates that the data behaves as sleep science predicts.

Limitations. The multi-modal label distributions caused by hard quality-class stratification prevent KS test convergence to Gaussian references. A softer stratification (continuous quality parameter) would produce more Gaussian-looking labels at the cost of reduced class balance. The sleep score formula is an approximation; real subjective quality ratings have additional random variability that is partially modelled by our residual noise but not perfectly reproduced.

Future work. Adding humidity-specific health effects (respiratory disruption at extreme RH), physiological signals (heart rate variability, movement), and a time-varying Poisson process (more disturbances in light sleep phases) would further increase realism. Validation against an actual bedroom sensor dataset would replace the Gaussian reference approximation.

6 Conclusion

We built a complete synthetic sleep environment dataset generator that produces 2,500 physically plausible, statistically validated sessions pairing bedroom environmental time-series with sleep quality labels. The pipeline applies four signal-processing techniques rigorously, and an ablation study quantifies each component's contribution. The Random Forest achieves R^2=0.795 on the test set, substantially above both the mean baseline and Ridge regression, confirming the non-linear structure introduced by our SP pipeline. All six sleep science sanity checks pass, demonstrating that the data correctly encodes known environment-sleep relationships. The complete reproducible pipeline runs in under 31 seconds from a clean install.

Contributions. Rushav Dash designed and implemented the signal processing pipeline (Butterworth filter, FFT analysis, pink noise synthesis, Poisson modelling) and the three-tier validation framework. Lisa Li designed and implemented the feature extraction layer, the ML pipeline (Random Forest, baselines, hyperparameter search), the ablation study, and all visualisation and documentation.

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